

Netflix Dataset: Data Cleaning and Exploratory Data Analysis

Summary Report

This project involved cleaning and analyzing a Netflix dataset containing 8,807 titles, including movies and TV shows. The objective was to improve data quality, perform exploratory data analysis (EDA), and uncover meaningful insights about Netflix's content library.

Cleaning Challenges Encountered

Several data quality issues were identified during the cleaning process:

- Missing values were present in the `director`, `cast`, `country`, `date_added`, `rating`, and `duration` columns.
- The `date_added` column was stored as a text field and required conversion to a datetime format.
- Text fields contained formatting inconsistencies that required standardization.
- The `duration` column contained mixed formats representing either minutes (movies) or seasons (TV shows).
- A duplicate records check was performed, and no duplicate entries were found.

To address these issues:

- Missing values in text-based columns were filled with `"Unknown"`.
- Missing values in `country` and `rating` were replaced using the mode.
- Date fields were converted to datetime format.
- Text data was standardized for consistency.
- The duration field was cleaned and prepared for analysis.

Key EDA Findings

The exploratory analysis revealed several notable patterns:

- Movies make up the majority of Netflix's content catalog compared to TV shows.
- The number of titles added to Netflix increased significantly over the years, particularly after 2015.
- A small number of countries contribute a large proportion of the available content.
- Certain content ratings appear far more frequently than others, indicating Netflix's focus on specific audience groups.
- Drama-related and international content categories are among the most common genres on the platform.

Top Insights

1. Netflix's catalog is dominated by movies, suggesting a stronger emphasis on film content than television series.
2. Content acquisition and production grew rapidly during the late 2010s, reflecting Netflix's global expansion strategy.
3. A few countries, particularly major entertainment-producing nations, account for a substantial share of Netflix content.
4. Mature and teen-oriented ratings are among the most common, highlighting Netflix's focus on broad audience segments.
5. Drama, international content, and documentary-related genres are highly represented, indicating strong viewer demand for diverse storytelling.

Conclusion

The dataset was successfully cleaned and standardized, enabling reliable analysis. The findings indicate that Netflix has experienced substantial growth in content additions, maintains a movie-heavy catalog, and offers a diverse range of genres and international productions. These insights provide a better understanding of Netflix's content strategy and audience focus.

Before submitting, update any findings that depend on your actual charts (for example, the exact top countries, ratings, or genres) so the report matches your results exactly. 🇮🇹 ✨